

Sukai Huang

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EXPERIENCE

Postdoctoral Research Fellow

Jun 2025 - Present

Monash University, Australia

- Researched on Neuro-symbolic systems for robotics reasoning and planning; Large language models for embodied AI.
- DARPA HARNESS Project (Hierarchical Abstractions and Reasoning for Neuro-Symbolic Systems): Developed neurosymbolic planning and granularity-aware instruction following modules for autonomous robotic systems, integrating perception, reasoning, and planning through neuro-symbolic frameworks.
- ONR SEA-AI Project (Neuro-Symbolic Enhanced Autonomy for Maritime Scene Understanding): Led mission analysis from natural language, world model construction, active perception, and sequential decision-making components for maritime autonomous systems.

Machine Learning Engineer

May 2022 - Nov 2023

FINVISE London, Remote Contract

- Implemented an Automated Valuation Model (AVM), enhancing property valuation accuracy to exceed 80%
- Automated data collection and preprocessing pipeline using multiprocessing Selenium and Pandas, efficiently handling over 20 million multimodal property datasets (textual, numerical, and categorical)
- Proposed the use of a Masked Autoencoder (MAE) as the model backbone, addressing the challenge of approximately 20% missing data in the dataset through advanced mask learning techniques

Software Developer

Oct 2023 - Nov 2023

Self-initiated Web App Project, <https://banpick.win/>

- Designed and constructed "BanPick", a dynamic game drafting tool tailored for DOTA2 enthusiasts, utilizing a parallel alpha-beta pruning algorithm, enhancing strategic planning and game experience

Tutor

Feb 2020 - Jul 2021

Australian National University, Canberra, ACT

- Planned and delivered interactive tutorials between 2017 and 2020, teaching Python and Data Management

EDUCATION

Doctor of Philosophy - Engineering and IT

Jul 2021 - Oct 2025

The University of Melbourne

- Thesis: Integrating Natural Language for Sequential Decision Problems
- Melbourne Research Scholarship for PhD
- Advisor: A.Prof. Nir Lipovetzky and Prof. Trevor Cohn

Bachelor of Advanced Computing (Honours)

Jul 2017 - Jul 2021

Australian National University

- Specialized in Machine Learning Systems
- Conducted research on Procedural Content Generation (PCG) using Generative Adversarial Networks
- First Class Honours; Chancellor's Letters of Commendation 2018, 2020; ANU Dean's Award 2019; GPA: 6.84 / 7.00

TECHNICAL SKILLS

- Programming: proficient in Python, with expertise in the Pytorch, Selenium, Numpy, Pandas, Langchain deepagents and Streamlit, complemented by relevant tools like Git, Hydra, Prefect, Terraform, Kedro and Wandb
- Past experience with Java, SQL, C++ and Haskell

AWARDS AND RECOGNITIONS

- Reviewer for ICAPS, ACL Rolling Review (ARR), and the Artificial Intelligence journal.
- Presentation at Kingston AI Symposium 2026 @ Adelaide, Australia
- AAAI 2025 Scholarship, Association for the Advancement of Artificial Intelligence
- ICAPS 2025 Scholarship, International Conference on Automated Planning and Scheduling
- Two oral presentations at Workshop on Planning in the Era of LLMs (LM4Plan @ AAAI 2025)
- Winner in the Aesthetic Track of the Angry Birds Level Generation Competition at IEEE COG'20
- Researcher in ANU Summer Scholarship Program 2019, working on emotion recognition model

SELECTED PUBLICATIONS

- Liu, Chunhua, Kabir Manandhar Shrestha, and **Sukai Huang**. "ALIGN: Word Association Learning for Cultural Alignment in Large Language Models." Annual Meeting of the Association for Computational Linguistics (ACL), 2026.
- De Deyne Simon, **Sukai Huang**, Kabir Manandhar Shrestha, Lea Frermann, and Chunhua Liu. "Fine-tuned Large Language Models predict human word associations and improve performance across lexical and semantic processing tasks." Annual Meeting of the Cognitive Science Society (CogSci), 2026.
- Cai, Zhixi, Fucai Ke, Kevin Leo, **Sukai Huang**, Maria Garcia de la Banda, Peter J. Stuckey, and Hamid RezaTofighi. "MATA: A Trainable Hierarchical Automaton System for Multi-Agent Visual Reasoning." International Conference on Learning Representations (ICLR), 2026.
- **Sukai Huang**, Zhang, Chenyuan, Hamid RezaTofighi, Mor Vered, and Buser Say. "Neurosymbolic Active Goal Recognition in Partially Observable Environments." International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS), 2026.
- **Sukai Huang**, Shu-Wei Liu, Nir Lipovetzky, and Trevor Cohn. "The Dark Side of Rich Rewards: Understanding and Mitigating Noise in VLM Rewards." ICAPS 2025 Workshop Briding the Gap Between AI Planning and RL (PRL)
- **Sukai Huang**, Nir Lipovetzky, and Trevor Cohn. "Chasing Progress, Not Perfection: Revisiting Strategies for End-to-End LLM Plan Generation." International Conference on Automated Planning and Scheduling (ICAPS), 2025 (Oral)
- **Sukai Huang**, Nir Lipovetzky, and Trevor Cohn. "Planning in the Dark: LLM-Symbolic Planning Pipeline without Experts." AAAI Conference on Artificial Intelligence (AAAI), 2025